

Pearson Edexcel International GCSE

# Mathematics A

## Modular Unit 1

### June 2022 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

29 adaptive question sections    100 marks

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Selected from Linear Higher Papers 1H and 2H

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**L-to-M | CONVERTED MODULAR EDITION**

Selected from Linear Higher papers; not an original Modular examination paper.

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## About this reconstructed paper

Each question section keeps the exact source demand beside practical working space. Source pixels are retained once; long source panels continue on numbered pages.

**Ownership rule:** Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. This book contains the demands owned by Unit 1; the selection contains 100 marks.

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| No. | Source                      | Chapter                                                                         | Focus                                                                                      | Marks | Page |
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| 2   | 4MA1-1H Q5 Q5(a)            | Expanding Brackets                                                              | Standard double bracket                                                                    | 2     | 6    |
| 3   | 4MA1-1H Q5 Q5(b)            | Solving Linear Equations                                                        | Linear unknown on both sides                                                               | 3     | 7    |
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| 7   | 4MA1-1H Q11 Q11(a)          | Factorising                                                                     | Difference of two squares                                                                  | 2     | 11   |
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| 9   | 4MA1-1H Q12 Q12(a), Q12(b)  | Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability | Construct a tree from conditional wording / Use branch-dependent conditional probabilities | 4     | 13   |
| 10  | 4MA1-1H Q15 Q15             | Combined & Conditional Probability                                              | Find exactly one, same or different independent outcomes                                   | 4     | 14   |
| 11  | 4MA1-1H Q16 Q16             | Surds                                                                           | Add or subtract like surds                                                                 | 3     | 15   |
| 12  | 4MA1-1H Q18 Q18             | Sine, Cosine Rule & Area of Triangles                                           | Find an acute angle using the sine rule                                                    | 5     | 16   |
| 13  | 4MA1-1H Q19 Q19             | Circles, Arcs & Sectors                                                         | Solve an algebraic circle-measure constraint                                               | 5     | 18   |
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| 20  | 4MA1-2H Q7 Q7               | Standard & Compound Units                                                       | Convert compound speed units                                                               | 3     | 27   |
| 21  | 4MA1-2H Q11 Q11(i), Q11(ii) | Factorising; Solving Quadratic Equations                                        | Negative constant monic quadratic / Monic quadratic roots by factors                       | 3     | 28   |
| 22  | 4MA1-2H Q14 Q14(a)          | Expanding Brackets                                                              | Three linear brackets                                                                      | 3     | 29   |
| 23  | 4MA1-2H Q15 Q15(a)          | Solving Linear Equations                                                        | Several fractional linear terms                                                            | 4     | 30   |

| No. | Source                                                                  | Chapter                                                                               | Focus                                                                                                                                                | Marks | Page |
|-----|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| 24  | 4MA1-2H Q16<br>Q16(a), Q16(b)(i),<br>Q16(b)(ii),<br>Q16(b)(iii), Q16(c) | Probability Diagrams - Venn<br>& Tree Diagrams; Combined<br>& Conditional Probability | Complete a three-set<br>Venn diagram / Read<br>regions and<br>probabilities from a<br>Venn diagram /<br>Calculate a direct<br>given-that probability | 8     | 31   |
| 25  | 4MA1-2H Q18 Q18                                                         | Rounding, Estimation &<br>Bounds                                                      | Algebraic expression<br>upper or lower bound                                                                                                         | 3     | 33   |
| 26  | 4MA1-2H Q19 Q19                                                         | Algebraic Fractions                                                                   | Substitute a variable<br>relation into a fraction                                                                                                    | 3     | 34   |
| 27  | 4MA1-2H Q20 Q20                                                         | Circles, Arcs & Sectors                                                               | Find annular,<br>overlapping or<br>multi-circle area                                                                                                 | 4     | 35   |
| 28  | 4MA1-2H Q22 Q22                                                         | Coordinate Geometry                                                                   | Kite or symmetric<br>vertex coordinates                                                                                                              | 7     | 37   |
| 29  | 4MA1-2H Q24 Q24                                                         | Algebraic Roots & Indices                                                             | Express one exponent<br>variable in terms of<br>others                                                                                               | 3     | 38   |

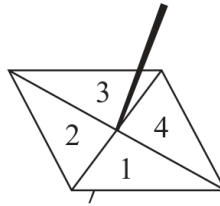
**Question 01**

4MA1-1H Q2 Q2

MODULAR UNIT 1

4 marks

- 2 Here is a biased 4-sided spinner.



The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

| Number      | 1    | 2 | 3    | 4 |
|-------------|------|---|------|---|
| Probability | 0.26 |   | 0.18 |   |

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Ravina is going to spin the spinner a number of times.

Ravina works out that an estimate for the number of times the spinner will land on 3 is 45

Work out an estimate for the number of times the spinner will land on 4

## Working space for Question 01

Continue your method

UNIT 1  
4 marks

WORKING SPACE

**Question 02**

4MA1-1H Q5 Q5(a)

MODULAR UNIT 1

2 marks

5 (a) Expand and simplify  $(n - 6)(n + 4)$ 

---

  
(2)

WORKING SPACE

**Question 03**

4MA1-1H Q5 Q5(b)

MODULAR UNIT 1

3 marks

(b) Solve  $2x - 3 = \frac{3x - 5}{4}$

Show clear algebraic working.

$x =$  .....  
(3)

WORKING SPACE



**Question 04**

4MA1-1H Q9 Q9(a)

MODULAR UNIT 1

1 marks

9 (a) Simplify  $8 \times (4t)^0$ 

(1)

WORKING SPACE

**Question 05**

4MA1-1H Q9 Q9(b)

MODULAR UNIT 1

1 marks

$$x^6 \div x^{-5} = x^p$$

(b) Find the value of  $p$  $p = \dots\dots\dots$   
(1)

WORKING SPACE

**Question 06**

4MA1-1H Q9 Q9(c)

MODULAR UNIT 1

2 marks

(c) Simplify fully  $(2k^2m^4)^3$ 

---

  
(2)

WORKING SPACE

**Question 07**

4MA1-1H Q11 Q11(a)

MODULAR UNIT 1

2 marks

11 (a) Factorise  $9x^2 - 4y^2$ 

(2)

WORKING SPACE

**Question 08**

4MA1-1H Q11 Q11(b)

MODULAR UNIT 1

3 marks

(b) Express  $\frac{7}{8} - \frac{x+3}{4x}$  as a single fraction in its simplest form.

.....  
(3)

WORKING SPACE

**Question 09**

4MA1-1H Q12 Q12(a), Q12(b)

MODULAR UNIT 1

4 marks

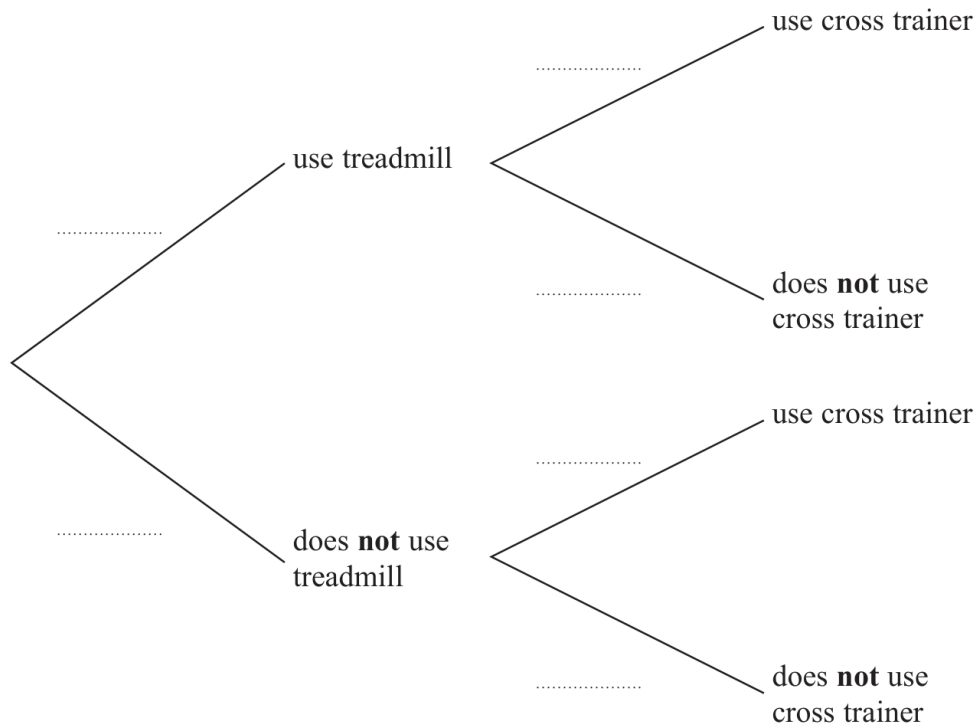
**12** Rudolf goes to the gym.

The probability that he will use the treadmill is 0.8

When he uses the treadmill, the probability that he will use the cross trainer is 0.3

When he does **not** use the treadmill, the probability that he will use the cross trainer is 0.6

(a) Complete the probability tree diagram for this information.



(2)

(b) Work out the probability that Rudolf uses both the treadmill and the cross trainer.

.....  
(2)

**Question 10**

4MA1-1H Q15 Q15

MODULAR UNIT 1

4 marks

- 15 Abraham is going to play a computer game.  
Abraham can win the game, draw the game or lose the game.

For any game that Abraham plays

the probability that he wins the game is 0.3  
the probability that he draws the game is 0.5  
the probability that he loses the game is 0.2

When Abraham wins a game, he scores +10 points.

When Abraham draws a game, he scores 0 points.

When Abraham loses a game, he scores -5 points.

Abraham plays 3 games and the points he scores in each of the 3 games are added together to get his total score.

Work out the probability that when he has played 3 games his total score is 0 points.

WORKING SPACE

**Question 11**

4MA1-1H Q16 Q16

MODULAR UNIT 1

3 marks

- 16 Without using a calculator, show that  $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$

Show your working clearly.

WORKING SPACE



**Question 12**

4MA1-1H Q18 Q18

MODULAR UNIT 1

5 marks

18 Here is triangle  $ABC$

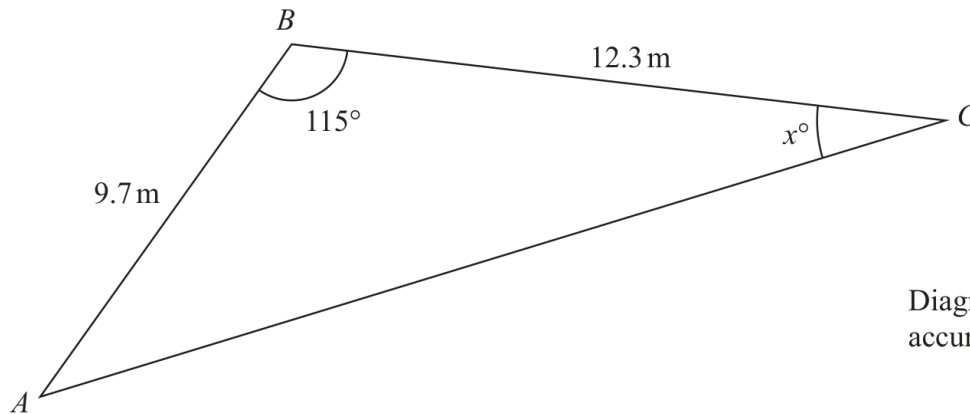


Diagram **NOT**  
accurately drawn

Work out the value of  $x$   
Give your answer correct to 3 significant figures.

$x =$  .....

## Working space for Question 12

Continue your method

UNIT 1

5 marks

WORKING SPACE

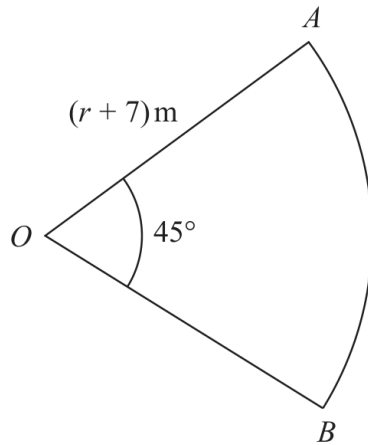
**Question 13**

4MA1-1H Q19 Q19

MODULAR UNIT 1

5 marks

19

Diagram **NOT**  
accurately drawn

$OAB$  is a sector  $S$  of a circle with centre  $O$  and radius  $(r + 7)$  metres.  
Angle  $AOB = 45^\circ$

A circle  $C$  has radius  $(r - 2)$  metres.

The area of sector  $S$  is twice the area of circle  $C$

Find the value of  $r$

Show your working clearly.

**Question 19 continued.** $r = \dots\dots\dots$

## Working space for Question 13

Continue your method

UNIT 1  
5 marks

WORKING SPACE

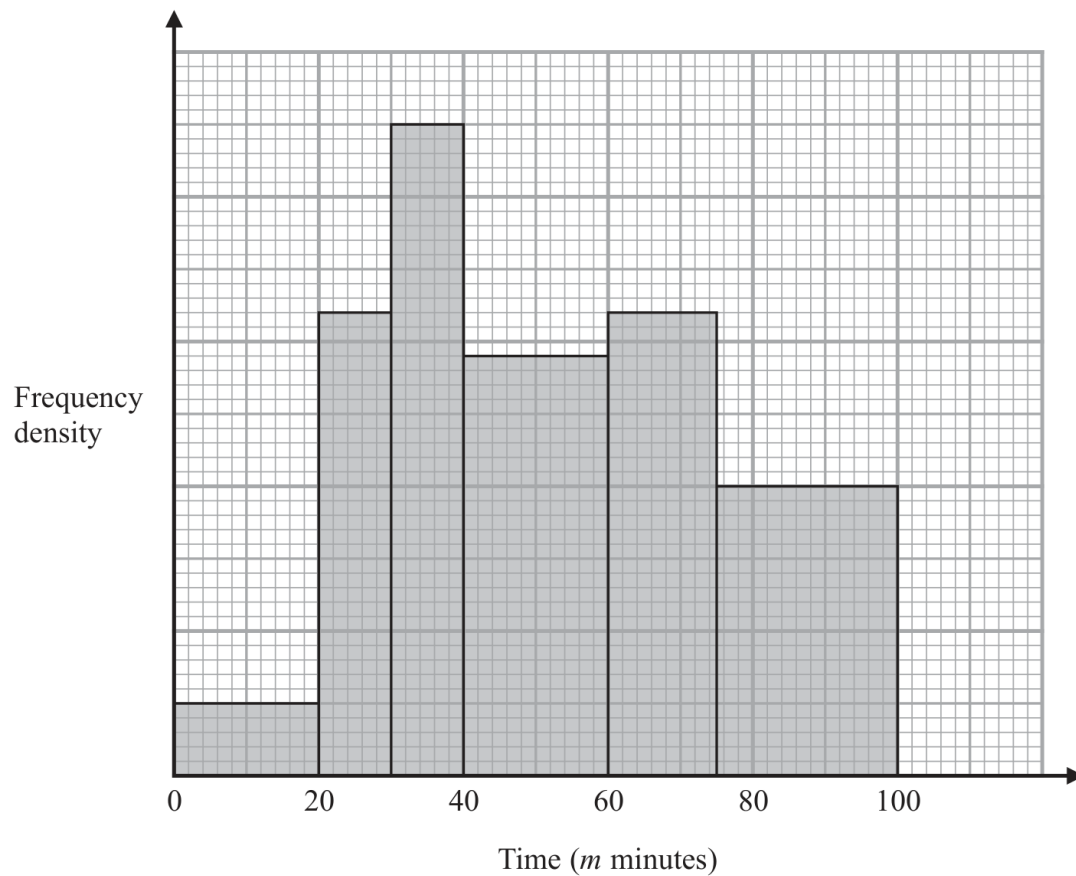
**Question 14**

4MA1-1H Q21 Q21

MODULAR UNIT 1

3 marks

- 21 The histogram shows information about the total time,  $m$  minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom  $m > 100$

There are 10 children for whom  $m \leq 20$

Work out an estimate for the number of children for whom  $50 < m \leq 80$

**Question 15**

4MA1-1H Q24 Q24

MODULAR UNIT 1

4 marks

24 Express each of  $a$ ,  $b$  and  $c$  in terms of  $q$  so that

$$q + 12x - qx^2$$

can be written as  $a - b(x - c)^2$

 $a = \dots\dots\dots$  $b = \dots\dots\dots$  $c = \dots\dots\dots$ 

WORKING SPACE

**Question 16**

4MA1-2H Q3 Q3

MODULAR UNIT 1

3 marks

- 3 An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.

Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

WORKING SPACE

**Question 17**

4MA1-2H Q4 Q4

MODULAR UNIT 1

3 marks

4 Show that  $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

WORKING SPACE



**Question 18**

4MA1-2H Q5 Q5

MODULAR UNIT 1

4 marks

- 5 The diagram shows an 8-sided shape  $ABCDEFGH$ .

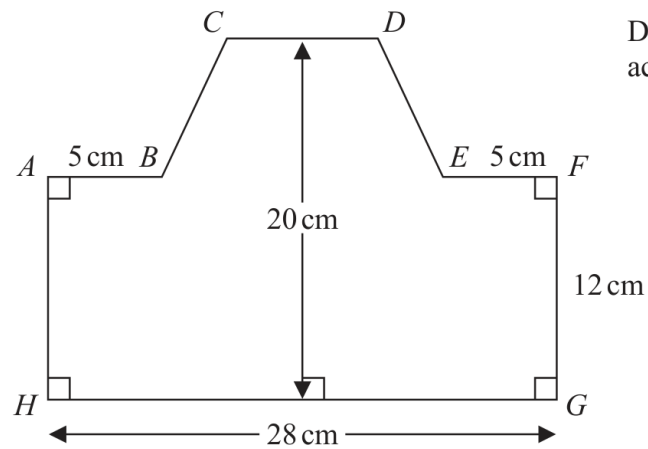


Diagram **NOT**  
accurately drawn

$$HG = 28 \text{ cm} \quad FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

$CD$  is parallel to  $HG$

The area of shape  $ABCDEFGH$  is  $434 \text{ cm}^2$

Find the length of  $CD$ .

..... cm

## Working space for Question 18

Continue your method

UNIT 1  
4 marks

WORKING SPACE

**Question 19**

4MA1-2H Q6 Q6

MODULAR UNIT 1

3 marks

- 6 The diagram shows triangle  $PQR$ .

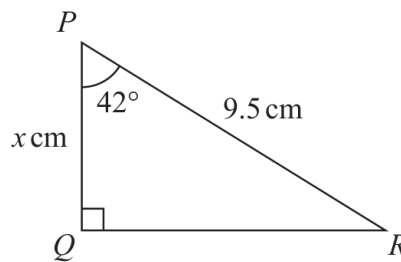


Diagram **NOT**  
accurately drawn

Work out the value of  $x$   
Give your answer correct to one decimal place.

$x =$  .....

WORKING SPACE

**Question 20**

4MA1-2H Q7 Q7

MODULAR UNIT 1

3 marks

7 Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

WORKING SPACE

**Question 21**

4MA1-2H Q11 Q11(i), Q11(ii)

MODULAR UNIT 1

3 marks

11 (i) Factorise  $x^2 + 5x - 24$ 

(2)

(ii) Hence, solve  $x^2 + 5x - 24 = 0$ 

(1)

WORKING SPACE

**Question 22**

4MA1-2H Q14 Q14(a)

MODULAR UNIT 1

3 marks

- 14 (a) Expand and simplify  $(5 - x)(2x + 3)(x + 4)$   
Show your working clearly.

---

(3)

WORKING SPACE

**Question 23**

4MA1-2H Q15 Q15(a)

MODULAR UNIT 1

4 marks

15 (a) Solve  $\frac{4x + 5}{3} - \frac{3 - 2x}{2} = 13$

Show clear algebraic working.

$x =$  .....  
(4)

WORKING SPACE

**Question 24**

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

MODULAR UNIT 1

8 marks

**16** 100 farmers are asked if they have goats ( $G$ ), sheep ( $S$ ) or chickens ( $C$ ) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

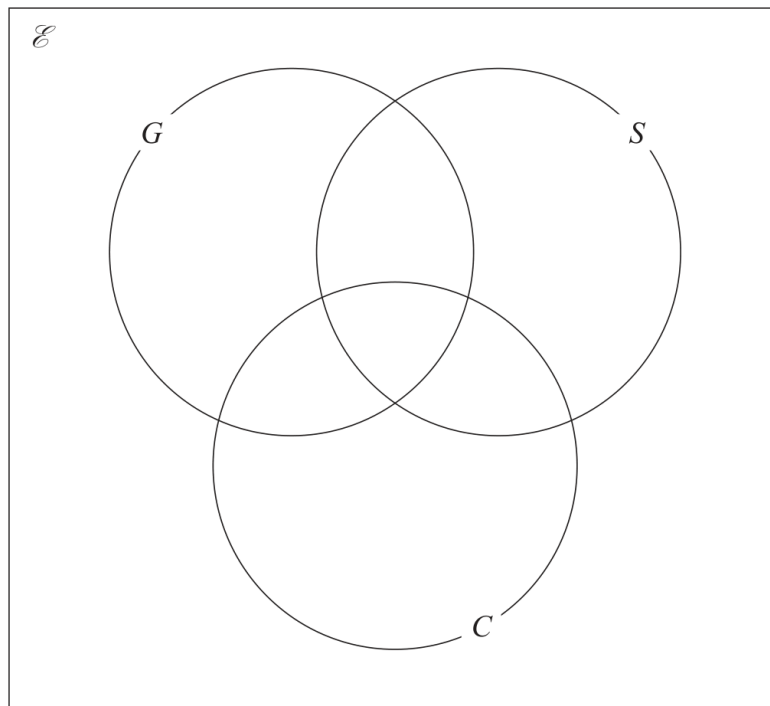
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

(b) Find



**Question 24 (continued)**

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

MODULAR UNIT 1

8 marks

(i)  $n(G)$

.....  
(1)

(ii)  $n([G \cup S]')$

.....  
(1)

(iii)  $n(G' \cap C)$

.....  
(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

.....  
(2)

**Question 25**

4MA1-2H Q18 Q18

MODULAR UNIT 1

3 marks

18  $X = \frac{2a - b}{f}$

$a = 7.5$  correct to 1 decimal place.

$b = 3.42$  correct to 2 decimal places.

$f = 2$  correct to the nearest whole number.

Work out the upper bound of the value of  $X$

Show your working clearly.

WORKING SPACE

**Question 26**

4MA1-2H Q19 Q19

MODULAR UNIT 1

3 marks

$$19 \quad a = \frac{14}{3x - 7} \quad x = \frac{7}{4y - 3}$$

Express  $a$  in the form  $\frac{py + q}{ry + s}$  where  $p, q, r$  and  $s$  are integers.

Give your answer in its simplest form.

 $a = \dots\dots\dots$ 

WORKING SPACE

**Question 27**

4MA1-2H Q20 Q20

MODULAR UNIT 1

4 marks

- 20 The diagram shows four identical circles drawn inside a square.

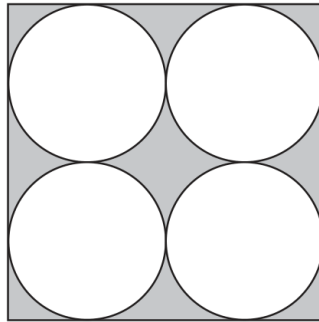


Diagram **NOT**  
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of  $40 \text{ cm}^2$

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

## Working space for Question 27

Continue your method

UNIT 1  
4 marks

WORKING SPACE

**Question 28**

4MA1-2H Q22 Q22

MODULAR UNIT 1

7 marks

- 22  $ABCD$  is a kite, with diagonals  $AC$  and  $BD$ , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

$A$  is the point with coordinates  $(-3, 4)$

The diagonals of the kite intersect at the point  $M$  with coordinates  $(0, 2)$

Given that  $AB = AD = 6.5$  cm and the  $x$  coordinate of  $B$  is positive,

find the coordinates of the points  $B$  and  $D$ .

(....., .....)

(....., .....)

WORKING SPACE

**Question 29**

4MA1-2H Q24 Q24

MODULAR UNIT 1

3 marks

24

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express  $x$  in terms of  $n$ 

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$ 

WORKING SPACE